

MERRA-2 and MERRAclim-2 variables in the extended dataset.

MERRA-2 (M2) Variables*

albedo	Surface albedo
cldtot	Total cloud area fraction
echange	Total land energy change per unit time (W/m2)
eflux	Positive latent heat flux (W/m2)
evland	Evaporation from land ((kg/m2)/s)
evpintr	Interception loss energy flux (W/m2)
evpsoil	Bare soil evaporation energy flux (W/m2)
evptrns	Transpiration energy flux (W/m2)
frwlt	Fractional wilting area
ghland	Downward heat flux into topsoil layer (W/m2)
grn	Vegetation greenness fraction (LAI-weighted)
gwetprof	Average profile soil wetness
gwetroot	Root zone soil wetness
hflux	Positive sensible heat flux (W/m2)
lai	Leaf area index
lwgnt	Surface net downward longwave flux (W/m2)
prectotcorr	Total observation-corrected surface precipitation ((kg/m2)/s)
prevtot	Total re-evaporation/sublimation of precipitation ((kg/m2)/s)
prmc	Total profile soil moisture content (m3/m3)
qinfil	Soil water infiltration rate ((km/m2)/s))
qv2m	2-meter specific humidity (kg/kg)
rzmc	Root zone soil moisture content (m3/m3)
speed	Surface wind speed (m/s)
swgnt	Surface net downward shortwave flux (W/m2)
t2m	2-meter air temperature (K)
tautot	Optical thickness of all clouds
ts	Surface skin temperature (K)
tsat	Surface temperature of saturated zone (K)
tsurf	Mean land surface temperature (K)
wchange	Total land water change per unit time ((kg/m2)/s)

MERRAclim-2 (MC) Variables

bio1	Annual mean temperature (C)
bio2	Mean diurnal temperature range (C)
bio3	Isothermality [(MC_Bio02/MC_Bio07)*100] (%)
bio4	Temperature seasonality [(Standard deviation*100)] (C)
bio5	Maximum temperature of the warmest month (C)
bio6	Minimum temperature of the coldest month (C)
bio7	Temperature annual range (MC_Bio05-MC_Bio06) (C)
bio8	Mean temperature of the wettest quarter (C)
bio9	Mean temperature of the driest quarter (C)
bio10	Mean temperature of the warmest quarter (C)
bio11	Mean temperature of the coldest quarter (C)
bio12	Annual precipitation (mm)
bio13	Precipitation of the wettest month (mm)
bio14	Precipitation of the driest month (mm)
bio15	Precipitation seasonality (Coefficient of variation) (%)
bio16	Precipitation of the wettest quarter (mm)
bio17	Precipitation of the driest quarter (mm)
bio18	Precipitation of the warmest quarter (mm)
bio19	Precipitation of the coldest quarter (mm)

*Details about the MERRA-2 collection can be found in: Bosilovich, M.G., R. Lucchesi, and M. Suarez. 2016. "MERRA-2: File Specification." *GMAO Office Note 9* (Version 1.1): 1–73.